

E-LOCK: Automated Lockout/Tagout (LOTO) System

Integrating Wireless Relays and Biometric Authentication for Secure Electrical Isolation

1. Project Overview

E-LOCK is a modern, IoT-enabled Lockout/Tagout (LOTO) safety system designed for high-risk industrial environments. Traditional LOTO systems rely on manual management of physical padlocks and tags, which are highly susceptible to human error, procedural delays, and miscommunications.

E-LOCK addresses these systemic vulnerabilities by digitizing the authorization process and automating physical electrical isolation. By integrating **ESP32 microcontrollers**, **biometric scanners**, and **shunt trip circuit breakers**, the system ensures that machinery is completely de-energized and can only be restored after verified physical presence and digital clearance.

2. Key Features

- **Biometric Authorization & RBAC:** Industrial equipment is protected by non-transferable fingerprint verification using the AS608 scanner, with strict Role-Based Access Control separating Admin and Employee privileges.
 - **Transient-Filtered Failsafe:** If the LOTO relay faults, the **ZMPT101B Voltage Sensor** detects the anomaly and signals the **Time Delay Relay (TDR)** to isolate the load. Simultaneously, the **ESP32** energizes the shunt relay to trip the circuit breaker *before* the TDR releases the current.
 - **Mechanical Override (Shunt Trip):** The ultimate physical fail-safe. If the system commands a shutdown but power remains (e.g., due to a welded relay), the hardware-level feedback loop ensures the ESP32 forces the Main Circuit Breaker into the OFF position.
 - **High-Speed Wireless Response:** Uses **ESP-NOW** for instant hardware-to-hardware communication between nodes, bypassing the need for a local Wi-Fi router for critical safety commands.
 - **Live Dashboard & Auditing:** A centralized Next.js web interface tracks real-time isolation status and maintains an automated, tamper-proof audit trail in a **PostgreSQL** database.
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3. User Interface & Access Flow (Dashboard Logic)

The system enforces strict security right from the initial interaction. The entry point of the system immediately requests a **Fingerprint Scan**. Based on the scanned identity, the system routes the user dynamically:

A. Admin / Supervisor Flow

If an **Admin** fingerprint is detected, the dashboard provides oversight and management tools:

1. **Logs (Monitor):** View the complete, real-time audit trail of all LOTO events.
2. **Registration (Input New Worker's Data):** Secure portal to register new employees into the system. The Admin registers the new user by inputting:
 - Employee Name
 - Secure PIN
 - Fingerprint Data

B. Employee Flow

If an **Employee** fingerprint is detected, the dashboard restricts access to operational safety tools:

1. **Maintenance:** Initiates the LOTO sequence to de-energize specific machinery for safe servicing.
2. **Monitoring:** Displays the real-time electrical status of the assigned machinery.

C. Fallback Security Mechanism

For *any* action requiring a fingerprint scan (initial login, accessing maintenance, or registering a worker):

- **3-Attempt Limit:** If the fingerprint scan fails or errors out **3 times** consecutively, the system will lock the scanner and request the user's **Registered PIN** as a manual fallback to proceed.
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4. Technology Stack

Web Dashboard (FSD Architecture)

- **Framework:** Next.js (App Router)
- **UI/UX:** Shadcn UI & TailwindCSS
- **Database:** PostgreSQL
- **ORM:** Drizzle ORM
- **Communication:** MQTT (Hardware to Server) / WebSockets (Server to Client)

Firmware & Embedded

- **Environment:** PlatformIO (VS Code)
 - **Microcontrollers:** Dual ESP32 Setup (Gateway & Field Controller)
 - **Protocols:** ESP-NOW (Node-to-Node) and Wi-Fi/MQTT (Node-to-Server)
 - **Language:** C++
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5. System Architecture (Hardware Nodes)

The system is divided into operational nodes to ensure redundancy and safety:

1. **The Gateway (ESP32 #1):** Handles the **AS608 Biometric Scanner**. Upon a successful scan, it checks the role (Admin/Employee), broadcasts a high-speed command via **ESP-NOW** to the Field Controller, and simultaneously logs the event to the **PostgreSQL** database.
 2. **The Field Controller (ESP32 #2):** Receives commands to toggle the control relays and physically cut power to the machinery.
 3. **The Feedback Loop & Failsafe:** The **ZMPT101B Voltage Sensor** continuously monitors the load side. If the system commands a shutdown but power remains, the **Time Delay Relay (TDR)** sends a signal to **ESP32 #2**. ESP32 #2 then triggers the **Shunt Trip** to mechanically trip the Main Circuit Breaker, ensuring ultimate load isolation.
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6. Project Structure (Feature-Sliced Design)

Plaintext

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E-LOCK/

└─ web/	# Next.js Web Dashboard
└─┬─ src/	
└─ app/	# App Router
└─ auth/	# PIN Fallback and Login handlers
└─ admin/	# Logs and New Worker Registration pages
└─ employee/	# Maintenance and Monitoring pages
└─ layout.tsx	# Root layout
└─ components/	# Shared UI (Shadcn primitives)
└─ config/	# Env vars & DB connections
└─ db/	# PostgreSQL Schema & Drizzle Migrations
└─ features/	# Domain-specific business logic
└─ ioto-status/	# Machine-side feedback & failsafes
└─ user-mgmt/	# RBAC, PIN fallback, & Biometric data
└─ logs/	# Historical audit trail & monitoring
└─ types/	# Shared TypeScript definitions
└─ firmware/	# Hardware Implementation (C++)
└─┬─ src/	
└─ main.cpp	# main.cpp
└─ gateway.cpp	# Biometric handling, RBAC routing, & Server comms
└─ field_controller.cpp	# Power toggling & Voltage monitoring
└─ failsafe.cpp	# ZMPT101B -> TDR -> Shunt trip logic
└─ lib/	# Hardware libraries
└─ platformio.ini	# Build configuration
└─ docs/	# Design, Project, & CAD Documentation

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7. Significance of the Study

- **Workforce Protection:** Prevents accidental start-ups by verifying the physical presence of authorized workers via non-transferable biometric data.
- **Safety Compliance:** Replaces error-prone manual logs with a real-time, automated dataset, assisting safety officers in workplace audits.
- **Future Innovation:** Establishes a foundational blueprint for **Industrial Internet of Things (IIoT)** safety infrastructures.